

Total No. of Printed Pages: 3

SUBJECT CODE NO: - RNP_8027
FACULTY OF SCIENCE AND TECHNOLOGY
F.E (Group B) (NEP) Sem - II
Examination November / December 2025 (To be held in February-2026)
Basics of Electronics Engineering

[Time: 3 Hours]

[Max. Marks: 60]

Please check whether you have got the right question paper.

N. B

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Q.nos 2, 3, 4, 5, 6 and 7

SECTION A

Q1

Solve the following questions

20

a) Which component allows current in one direction only?

- a. Resistor
- b. Capacitor
- c. Inductor
- d. Diode

b) FET is a _____ controlled device.

- a. Current
- b. Voltage
- c. Power
- d. Capacitance

c) Which of the following is an active component?

- a. Resistor
- b. Capacitor
- c. Inductor
- d. Transistor

d) An op-amp as a comparator compares:

- a. two currents
- b. two voltages
- c. two resistors
- d. two capacitors

- e) Which of these is a common op-amp IC?
- a. IC 555
 - b. IC 7805
 - c. IC 741
 - d. IC 4017
- f) Which gate gives low output when both inputs are high?
- a. AND
 - b. XOR
 - c. NOR
 - d. NAND
- g) Full adder adds:
- a. 1 bit
 - b. 2 bits
 - c. 3 bits
 - d. 4 bits
- h) Which transducer is used for displacement measurement?
- a. Thermistor
 - b. Strain Gauge
 - c. LVDT
 - d. RTD
- i) Transducers that do not require an external power source are:
- a. Active
 - b. Passive
 - c. Both A and B
 - d. None
- j) Which of the following is not a transmission medium?
- a. Coaxial cable
 - b. Optical fiber

- c. Air
- d. Generator

SECTION B

- | | | |
|------|---|-----------|
| Q.2 | a) Draw and Explain basic block diagram of DC Regulated power supply. | 05 |
| | b) Explain 3 terminal fixed regulators in detail | 05 |
| Q.3 | a) Write short note on Field Effect Transistor | 05 |
| | b) What are the different configurations of BJT explain in detail. | 05 |
| Q.4. | a) Explain different types of Number systems in detail with one example each. | 05 |
| | b) Explain Op-Amp as a differentiator and Integrator | 05 |
| Q.5. | a) Explain different types of Number systems in detail with one example each. | 05 |
| | b) Classify and Explain different types of logic gates. | 05 |
| Q.6 | a) Write short note on LVDT. | 05 |
| | b) Explain any two temperature measurement transducers | 05 |
| Q.7 | a) Draw and Explain the block diagram of different elements of communication system | 05 |
| | b) State and explain different transmission media | 05 |